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THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



TerminalFix ClickFix Reverse-Tunnel Intrusion Campaign Daily Priority Update

Threat Report

Classification	TLP:CLEAR
Published	2026-09-01
Version	1.0
Author	Lost Boys Cyber
Report Type	Threat Report

TLP:CLEAR | Lost Boys Cyber | TerminalFix ClickFix Reverse-Tunnel Intrusion Campaign Daily Priority Update - Threat Report

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1. Executive Summary

Microsoft Threat Intelligence reported a TerminalFix campaign on 2026-08-28. Public reporting describes TerminalFix as a ClickFix variant that uses fake CAPTCHA / Cloudflare-style verification prompts to convince users to run malicious PowerShell or Windows Terminal commands. Instead of only delivering commodity theft tooling, the activity is notable because it deploys reverse-tunnel capability that can provide attackers with internal network access.

This topic was selected for the daily general threat set because the behavior is immediately hutable in endpoint telemetry: user-launched PowerShell following browser activity, suspicious command lines copied from web prompts, DLL sideload/staging, image or PNG retrieval used for payload staging, and Python/client reverse-tunnel execution. The activity is relevant across sectors, including aviation and aerospace operators with heavy use of SaaS portals, third-party web workflows, and roaming endpoints.

Defender priority for 2026-09-01: educate service desks on fake verification prompts, hunt for suspicious PowerShell and Terminal use launched by browsers, monitor unexpected outbound tunnels/WebSocket-like traffic, and review egress controls on user endpoints.

2. Intelligence Requirements

Question	Current Answer	Confidence
Is this actively relevant to enterprise defenders this week?	Yes. Microsoft reported the campaign on 2026-08-28 and multiple outlets amplified it through 2026-08-31/09-01.	High
What makes it different from generic ClickFix?	Public reporting says TerminalFix directs users to Terminal/PowerShell and deploys reverse-tunnel access rather than only a simple downloader.	Medium-High
Are sector-specific victims confirmed?	Available snippets describe multi-industry exposure; this report treats sector relevance as exposure-based unless confirmed victimology appears in primary sources.	Medium

3. Source Review & Confidence

Source	Contribution	Confidence
Microsoft Threat Intelligence	Primary campaign analysis, attack-chain framing, detection/hunting guidance per search result metadata	High
BleepingComputer	Timely defender-facing corroboration and simplified campaign summary	Medium
The Hacker News	Secondary corroboration of fake CAPTCHA and reverse-tunnel claims	Medium
The Register / other reputable security coverage	Additional public context on PNG/staging and reverse tunnel reporting	Medium

4. Threat Activity Overview

TerminalFix follows the broader ClickFix social-engineering pattern: a compromised or malicious website presents a fake verification step and instructs the user to execute copied commands. The campaign raises risk because a successful endpoint compromise can become a network pivot when reverse-tunnel tooling is deployed.

Reported behaviors include fake Cloudflare CAPTCHA lures, PowerShell or Windows Terminal execution, staged payload retrieval, DLL sideloading or script-based staging in some writeups, and reverse-tunnel client deployment. Defenders should focus on the user-action bridge between browser and shell rather than waiting for a single malware hash.

5. Affected Sectors and Exposure

Sector / Environment	Exposure Rationale
Enterprise Windows fleets	User-executed PowerShell/Terminal commands are the core abuse path.
Aviation and aerospace operators	High reliance on web portals, remote work, travel platforms, and third-party supplier access increases lure exposure.
Critical infrastructure	Reverse tunnels can bypass simple inbound firewall assumptions and create internal access paths.
Managed service and support environments	Shared admin workflows can turn one user endpoint into broad lateral-access opportunity.

6. Tactics, Techniques, and Procedures

Tactic / Phase	Observed Technique	Evidence
Initial Access	T1189 Drive-by Compromise / social-engineering via compromised website	Microsoft and secondary reporting describe compromised/fake CAPTCHA pages.
Execution	T1059.001 PowerShell	Reports state users are instructed to run PowerShell/Windows Terminal commands.
Defense Evasion	T1027 Obfuscated Files or Information	Public snippets mention staged content and image/PNG use; validate locally before blocking on this alone.
Command and Control	T1090 Proxy / reverse tunnel	Campaign is centered on reverse-tunnel access.

7. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
Fake CAPTCHA / Cloudflare verification prompt	Lure pattern	Awareness, browser-to-shell hunt pivot
PowerShell launched from user context after browser activity	Host behavior	Primary detection signal
Reverse-tunnel client such as client.py or nonstandard tunnel binary	Tooling behavior	Hunt for unauthorized egress tunnel

Unexpected WebSocket/long-lived outbound sessions	Network behavior	Proxy/firewall analytics
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8. Detection Engineering

8.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Browser-spawned PowerShell or Terminal	EDR process telemetry	Catch fake verification command execution
Encoded/download cradle command lines	PowerShell logs, command-line telemetry	Identify staged payload retrieval
Python or unusual client process making long-lived outbound connections	EDR/network telemetry	Surface reverse-tunnel clients

8.2. Sigma / Detection Content

```

title: Browser Initiated PowerShell or Windows Terminal After Fake Verification Lure
status: experimental
description: Detects browser processes launching PowerShell, pwsh, cmd, or Windows Terminal, a common ClickFix/TerminalFix behavior.
logsource:
  product: windows
  category: process_creation
detection:
  parent_browser:
    ParentImage|endswith:
      - '\chrome.exe'
      - '\msedge.exe'
      - '\firefox.exe'
      - '\brave.exe'
  child_shell:
    Image|endswith:
      - '\powershell.exe'
      - '\pwsh.exe'
      - '\WindowsTerminal.exe'
      - '\cmd.exe'
  suspicious_args:
    CommandLine|contains:
      - 'Invoke-WebRequest'
      - 'iwr '
      - 'curl '
      - 'FromBase64String'
      - '-enc'
      - 'Start-Process'
  condition: parent_browser and child_shell and suspicious_args
fields:
  - UtcTime
  - User
  - ParentImage
  - Image
  - CommandLine
falsepositives:
  - Developer workflows and documented helpdesk scripts launched from browser portals
level: high

```

8.3. Hunt Query

```

DeviceProcessEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName in~ ('chrome.exe','msedge.exe','firefox.exe','brave.exe')
| where FileName in~ ('powershell.exe','pwsh.exe','cmd.exe','WindowsTerminal.exe')
| where ProcessCommandLine has_any ('Invoke-WebRequest','iwr ', 'curl ', 'FromBase64String','-

```

```
enc', 'Start-Process')
| project Timestamp, DeviceName, AccountName, InitiatingProcessFileName, FileName, ProcessCommandLine
| order by Timestamp desc
```

9. Threat Hunting

9.1. Hunt Hypothesis

TerminalFix compromises produce a close temporal sequence: browser activity, user-context shell execution, staged download or decoding, then persistent or long-lived outbound tunnel traffic.

9.2. Hunt Query

```
let Shells = DeviceProcessEvents
| where Timestamp > ago(14d)
| where FileName in~
('powershell.exe', 'pwsh.exe', 'cmd.exe', 'WindowsTerminal.exe', 'python.exe', 'pythonw.exe')
| where ProcessCommandLine has_any ('iwr', 'Invoke-
WebRequest', 'curl', 'http', 'ws://', 'wss://', 'client.py', 'tunnel');
DeviceNetworkEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName in~
('powershell.exe', 'pwsh.exe', 'python.exe', 'pythonw.exe', 'cmd.exe')
| join kind=innerunique Shells on DeviceId
| where Timestamp between (Timestamp1 .. datetime_add('hour', 2, Timestamp1))
| summarize Connections=count(), RemoteIPs=dcount(RemoteIP), FirstSeen=min(Timestamp),
LastSeen=max(Timestamp) by DeviceName, InitiatingProcessFileName, InitiatingProcessCommandLine
| order by Connections desc
```

10. Risk Assessment

Dimension	Assessment
Likelihood	High for exposure to lures; medium for successful execution where PowerShell controls are mature.
Impact	High where reverse tunnels reach internal services, privileged workstations, or third-party access networks.
Confidence	Medium-high for the attack pattern; lower for sector-specific victimology in the sources available to this cron run.

11. Recommended Actions

Priority	Owner	Action
Critical	SOC	Deploy browser-to-shell and suspicious tunnel hunts.
High	Endpoint engineering	Enforce PowerShell constrained language, script block logging, and ASR rules where practical.
High	Network security	Alert on unauthorized outbound tunnels, wss/ws anomalies, and long-lived sessions from user endpoints.
Medium	Awareness / service desk	Warn users about fake CAPTCHA/verification prompts that ask them to run commands.

12. References

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